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GCR equation derivation EoR-FG: the systematics as gain case

Glossary of terms:

e: EoR component (a vector)

a_{fg} : Foreground component (a vector)

b_{sys} : Systematics component (a vector)

n: noise

H: The systematics template (defined as $H = \langle b_{sys} b_{sys}^\dagger \rangle$)

E: EoR covariance ($E = \langle ee^\dagger \rangle$)

G: Foregrounds covariance ($G = \langle a_{fg} a_{fg}^\dagger \rangle$)

N: Noise covariance matrix

Maths work-through:

Let us define $y = H @ b_{sys}$ as the systematics component. If we treat this component as the gain, the "signal" component becomes:

$$s = y(e + a_{fg})$$

Then, the covariances can be defined as:

$$S = \langle ss^\dagger \rangle = \langle y(e + a_{fg})(e + a_{fg})^\dagger y^\dagger \rangle = \langle yee^\dagger y^\dagger \rangle + \langle yea_{fg}^\dagger y^\dagger \rangle + \langle ya_{fg}e^\dagger y^\dagger \rangle + \langle ya_{fg}a_{fg}^\dagger y^\dagger \rangle$$

Since e and a_{fg} are independent in this scenario, we can neglect the cross terms and are left with:

$$S = \langle y e e^\dagger y^\dagger \rangle + \langle y a_{fg} a_{fg}^\dagger y^\dagger \rangle = y \langle e e^\dagger \rangle y^\dagger + y \langle a_{fg} a_{fg}^\dagger \rangle y^\dagger$$

$$\Rightarrow \boxed{S = y E y^\dagger + y G y^\dagger}$$

also, the complete data model (with noise) is defined as:

$$\boxed{d = s(e + a_{fg}) + n}$$

Now the Bayesian posterior is defined as:

$$p(s|S, N, d) \propto p(d|s, S, N) p(s|S)$$

After the Gaussian approximation, this becomes:

$$p(s|S, N, d) \propto \exp[-(d - s)^\dagger N^{-1} (d - s)] \exp[-s^\dagger S^{-1} s]$$

Expanding the prior, taking the log of the posterior, we can find the maxima of the above by doing:

$$\frac{\delta}{\delta e} \left[-(d - s)^\dagger N^{-1} (d - s) - e^\dagger y^\dagger E^{-1} y e - a_{fg}^\dagger y^\dagger G^{-1} y a_{fg} \right] = 0$$

and

$$\frac{\delta}{\delta a_{fg}} \left[-(d - s)^\dagger N^{-1} (d - s) - e^\dagger E^{-1} e - a_{fg}^\dagger G^{-1} a_{fg} \right] = 0$$

Expanding s to $s = y(e + a_{fg})$ and performing the above equations, we are left with this system of two equations:

1. $(y^\dagger N^{-1} y + y^\dagger E^{-1} y) e + (y^\dagger N^{-1} y G) a_{fg} = y^\dagger N^{-1} d$
2. $(G^\dagger y^\dagger N^{-1} y) e + (G^\dagger y^\dagger N^{-1} y G + y^\dagger G^{-1} y) a_{fg} = G^\dagger y^\dagger N^{-1} d$

Writing the matrix representation for this system gives us:

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